Estéban Gabory

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Appointments

2025 – Now
- **Postdoc** Wrocław, Poland,
  Supervisor: Paweł Gawrychowski

2024 – 2025
- **Assegno di Ricerca (Post-doctoral position)** Palermo, Italy,
  Supervisor: Gabriele Fici,
  Title: *Combinatorics on words for machine learning*.

Education

2021 – 2024
- **Ph.D. in Computer Science, CWI** Amsterdam, The Netherlands,
  Supervisors: Solon P. Pissis, Leen Stougie,
  Thesis title: *Variable strings for pangenomes: Matching, Comparison, Alignment*.

2020 – 2021
- **M.Sc. Fundamental Mathematics, Université de Paris** (Second year) Paris, France,
  Supervisor: Laurent Bulteau,
  Thesis title: *Parametrized algorithms for consensus problems with swaps*.

2020
- **Summer research internship, Université de Brest** Brest, France,
  Supervisor: Johannes Huisman,
  Topic: *Class Field Theory*.

2019 – 2020
- **M.Sc. Fundamental Mathematics, Université Rennes 1** (First year) Rennes, France,
  Supervisor: Ludovic Marquis,
  Thesis title: *Tessellations of the Hyperbolic Plane*.

2018–2019
- **Exchange year in the department of Mathematics, Sapienza Università di Roma** Rome, Italy.

2017 – 2019
- **Bachelor in Mathematics, Sorbonne Université** Paris, France.

Teaching

Università di Palermo

- Let's create an app (Editions in 2024 and 2025) - Introduction to JavaScript app creation for high school students

Vrije Universiteit

- Algorithms in Sequence Analysis, Master Course (2023) - **Teaching Assistant**.

Université de Paris

- Undergraduate Algebra, third year (2021) - **Teaching Assistant**.
- Undergraduate Algebra, second year (2021) - **Teaching Assistant**.
- Undergraduate Analysis, third year (2021) - **Teaching Assistant**.
- Undergraduate Analysis, second year (2021) - **Teaching Assistant**.

Association Science Technologie et Santé

- Introduction to programming with "Scratch" - Elementary school supervision for scientific activities.

Professional service

Program Committee Memberships

- International Symposium on String Processing and Information Retrieval (SPIRE 2026)
- IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2023).

Reviewing

- International Colloquium on Automata, Languages, and Programming (ICALP 2026) - **Subreviewer**.
- Symposium on Theory of Computing (STOC 2026) - **Subreviewer**.
- International Conference on Developments in Language Theory (DLT 2026) - **Subreviewer**.
- Prague Stringology Conference (PCS 2025) - **Subreviewer**.
- Annual Symposium on Combinatorial Pattern Matching (CPM 2025) - **Subreviewer** .
- Symposium on Experimental Algorithms (SEA 2025) - **Subreviewer** .
- Workshop on Algorithms in Bioinformatics (WABI 2024) - **Subreviewer**.
- Annual Symposium on Combinatorial Pattern Matching (CPM 2024) - **Subreviewer** .
- Symposium on String Processing and Information Retrieval (SPIRE 2023) - **Subreviewer**.
- Annual Symposium on Combinatorial Pattern Matching (CPM 2023) - **Subreviewer** .
- Workshop on Algorithms in Bioinformatics (WABI 2022) - **Subreviewer**.

Organising Committee Memberships

- Symposium on String Processing and Information Retrieval (SPIRE 2023).
- European Symposium on Algorithms (ALGO 2023).

PhD Courses, schools, and workshops

- Landelijk Netwerk Matematische Beslikunde - "Randomized Algorithms" (2023).
- Winter School - “Collaborating with wet-lab biologists” (2023).
- Landelijk Netwerk Matematische Beslikunde - "Networks and Semidefinite Programming" (2022).
- Annual ALPACA Workshop - “Pan-genome sequencing and analysis” (2022).
- Summer School - “Algorithmic techniques in pan-genomics” (2022).
- Landelijk Netwerk Matematische Beslikunde - "Integer Programming Methods" (2021).
- Annual ALPACA Workshop - “Challenges of computational pan-genomics” (2021).

Research Grants

- Marie Skłodowska-Curie Innovative Training Network (MSCA-ITN-2020) on ALgorithms for PAngenome Computational Analysis (ALPACA) – Role: Early Stage Researcher.

Research Publications

Journal Articles

[3] Gabriele Fici and Estéban Gabory, “Generalized De Bruijn Words, Invertible Necklaces, and the Burrows-Wheeler Transform,” *LIPIcs, Volume 345, MFCS 2025*, vol. 345, Paweł Gawrychowski, Filip Mazowiecki, and Michał Skrzypczak, Eds., 48:1–48:18, 2025, ISSN: 1868-8969. DOI: 10.4230/LIPICS.MFCS.2025.48. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.MFCS.2025.48.

[5] Estéban Gabory, Njagi Moses Mwaniki, Nadia Pisanti, Solon P. Pissis, Jakub Radoszewski, Michelle Sweering, and Wiktor Zuba, "Elastic-degenerate string comparison," *Inf. Comput.*, vol. 304, p. 105 296, 2025. DOI: 10.1016/J.IC.2025.105296. URL: https://doi.org/10.1016/j.ic.2025.105296.

[7] Giulia Bernardini, Estéban Gabory, Solon P. Pissis, Leen Stougie, Michelle Sweering, and Wiktor Zuba, "Elastic-degenerate string matching with 1 error or mismatch," *Theory of Computing Systems*, Sep. 2024, ISSN: 1433-0490. DOI: 10.1007/s00224-024-10194-8. URL: https://doi.org/10.1007/s00224-024-10194-8.

[9] Esteban Gabory, Moses Njagi Mwaniki, Nadia Pisanti, Solon P. Pissis, Jakub Radoszewski, Michelle Sweering, and Wiktor Zuba, "Pangenome comparison via ed strings," *Frontiers in Bioinformatics*, vol. 4, 2024, ISSN: 2673-7647. DOI: 10.3389/fbinf.2024.1397036. URL: https://www.frontiersin.org/journals/bioinformatics/articles/10.3389/fbinf.2024.1397036.

Conference Proceedings

[1] Pawel Gawrychowski Itai Boneh Estéban Gabory and Adam Górkiewicz, "Balancing Two-Dimensional Straight-Line Programs (to appear)," in *37th Annual Symposium on Combinatorial Pattern Matching (CPM 2026)*.

[6] Rocco Ascone, Giulia Bernardini, Alessio Conte, Massimo Equi, Estéban Gabory, Roberto Grossi, and Nadia Pisanti, "A Unifying Taxonomy of Pattern Matching in Degenerate Strings and Founder Graphs," in *24th International Workshop on Algorithms in Bioinformatics (WABI 2024)*, Solon P. Pissis and Wing-Kin Sung, Eds., ser. Leibniz International Proceedings in Informatics (LIPIcs), vol. 312, Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 14:1–14:21, ISBN: 978-3-95977-340-9. DOI: 10.4230/LIPIcs.WABI.2024.14. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.WABI.2024.14.

[8] Esteban Gabory, Chang Liu, Grigorios Loukides, Solon P. Pissis, and Wiktor Zuba, "Space-efficient indexes for uncertain strings," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 4828–4842. DOI: 10.1109/ICDE60146.2024.00367.

[10] Esteban Gabory, Moses Njagi Mwaniki, Nadia Pisanti, Solon P. Pissis, Jakub Radoszewski, Michelle Sweering, and Wiktor Zuba, "Comparing Elastic-Degenerate Strings: Algorithms, Lower Bounds, and Applications," in *34th Annual Symposium on Combinatorial Pattern Matching (CPM 2023)*, Laurent Bulteau and Zsuzsanna Lipták, Eds., ser. Leibniz International Proceedings in Informatics (LIPIcs), vol. 259, Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023, 11:1–11:20, ISBN: 978-3-95977-276-1. DOI: 10.4230/LIPIcs.CPM.2023.11. URL: https://drops.dagstuhl.de/opus/volltexte/2023/17965.

[11] Estéban Gabory, Eric Rivals, Michelle Sweering, Hilde Verbeek, and Pengfei Wang, "Periodicity of degenerate strings," in *Proc. of 26th Prague Stringology Conference*, Jan Holub and Solon Pissis, Eds., ISBN: 978-80-01-07206-6, Faculty of Information Technology, Czech Technical University in Prague, Prague, Czech Republic: Czech Technical University in Prague, Aug. 2023, pp. 42–56.

[12] Giulia Bernardini, Alessio Conte, Esteban Gabory, Roberto Grossi, Grigorios Loukides, Solon P Pissis, Giulia Punzi, and Michelle Sweering, "On Strings Having the Same Length-k Substrings," in *33rd Annual Symposium on Combinatorial Pattern Matching (CPM)*, Prague, Czech Republic, 2022. DOI: 10.4230/LIPIcs.CPM.2022.16. URL: https://inria.hal.science/hal-03829979.

[13] Giulia Bernardini, Esteban Gabory, Solon P. Pissis, Leen Stougie, Michelle Sweering, and Wiktor Zuba, "Elastic-degenerate string matching with 1 error," in *LATIN 2022: Theoretical Informatics*, Armando Castañeda and Francisco Rodríguez-Henríquez, Eds., Cham: Springer International Publishing, 2022, pp. 20–37, ISBN: 978-3-031-20624-5. DOI: 10.1007/978-3-031-20624-5_2.

Preprints

[2] Itai Boneh, Estéban Gabory, Paweł Gawrychowski, and Adam Górkiewicz, *Balancing Two-Dimensional Straight-Line Programs*, Nov. 27, 2025. DOI: 10.48550/arXiv.2511.22212. arXiv: 2511.22212 [cs]. URL: http://arxiv.org/abs/2511.22212, pre-published.

[4] Gabriele Fici, Estéban Gabory, Giuseppe Romana, and Marinella Sciortino, *Unclustered BWTs of any Length over Non-Binary Alphabets*, Aug. 28, 2025. DOI: 10.48550/arXiv.2508.20879. arXiv: 2508.20879 [cs]. URL: http://arxiv.org/abs/2508.20879, pre-published.